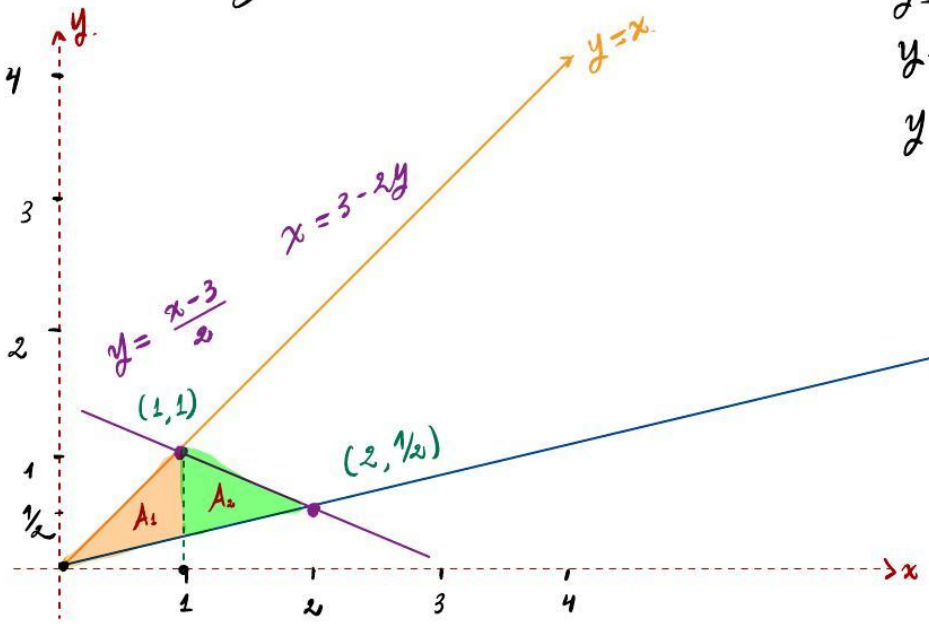
Problem for you: Find the area of the region enclosed by $x+2y=3$ $y=x$ and $y=\frac{1}{4}x$ using.

a). x as variable of integration

b). y as variable of integration.

Sln.



$y=x$

$y=\frac{1}{4}x$

$y=\frac{3-x}{2}$.

$y=\frac{1}{4}x$

$x=4y$.

a). x Variable:

$$A = A_1 + A_2 = \int_0^1 \{x - \tfrac{1}{4}x\}\,dx + \int_1^2 \{\tfrac{x-3}{2} - \tfrac{1}{4}x\}\,dx$$

(A_1: $x - \frac{1}{4}x$; A_2: $\frac{x-3}{2} - \frac{1}{4}x$)

$$= \int_0^1 \tfrac{3}{4}x\,dx + \int_1^2 \tfrac{x-6}{2}\,dx$$

$$= \tfrac{3}{8}x^2\Big|_0^1 + \tfrac{1}{2}\left\{\tfrac{x^2}{2} - 6x\Big|_1^2\right\} = \tfrac{3}{4}.$$

b). y Variable:

$$A = A_1 + A_2 = \int_0^{1/2} \{4y - y\}\,dy + \int_{1/2}^1 \{3 - 2y - y\}\,dy$$

$$= \int_0^{1/2} 3y\,dy + \int_{1/2}^1 (3-3y)\,dy$$

$$\int_a^b f(x)\,dx + \int_b^c f(x)\,dx = \int_a^c f(x)\,dx.$$

$$\overset{\times}{=} \int_0^1 \{3y + 3 - 3y\}\,dy.$$

this is wrong! ⚠

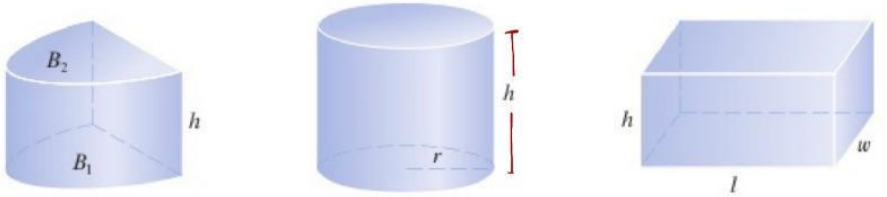
$$= \frac{3}{2}y^2\Big|_0^{1/2} + \left(3y - \frac{3}{2}y^2\Big|_{1/2}^{1}\right)$$

$$= \frac{3}{8} + \left(3 - \frac{3}{2} - \frac{3}{2} + \frac{3}{8}\right)$$

$$= \frac{3}{4}.$$

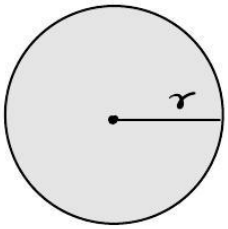
5.2 Volumes

We start with a simple type of solid called a cylinder.



(a) Cylinder $V = Ah$ (b) Circular cylinder $V = \pi r^2 h$ (c) Rectangular box $V = lwh$

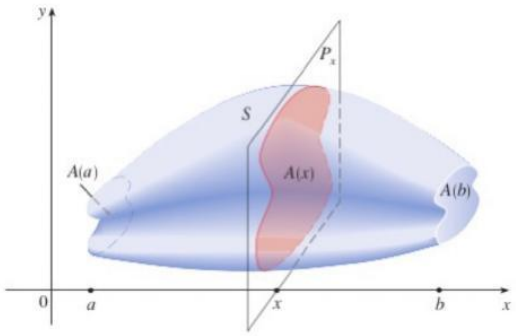
Rmk. Notice that in the case b). $V = \pi r^2 h$ we have a circle in the base.

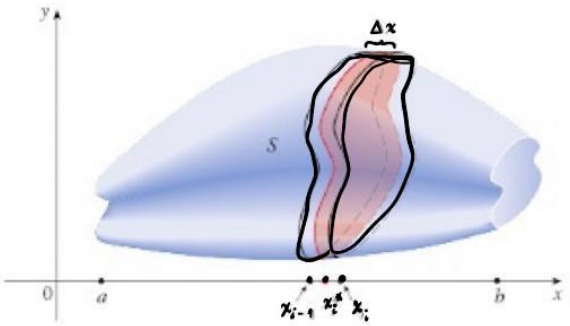


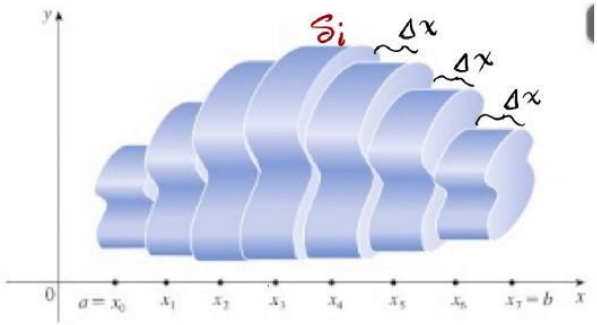
which area is πr^2.

We can define the volume V of a cylinder as $V = A \cdot h$.

Let's divide S into n "slabs" of equal width Δx by using planes $P_1, P_2, \ldots, P_n$ to slice the solid.







The volume of this cylinder is $A(x_i^*)\cdot\Delta x$, so an approx to our intuitive conception of the volume of the ith slab S_i is:

$$V(S_i) \approx A(x_i^*)\,\Delta x$$

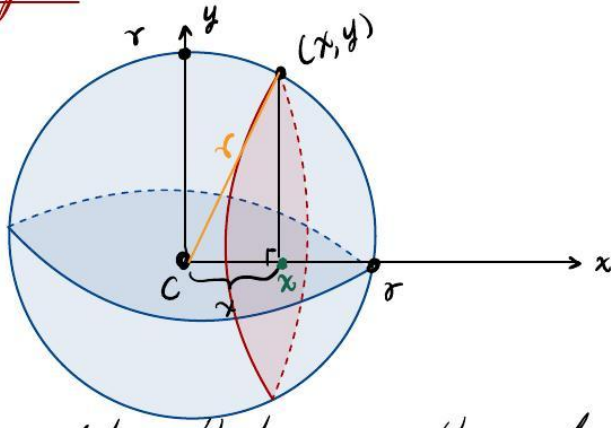
then $$V \approx \sum_{i=1}^{n} A(x_i^*)\,\Delta x$$

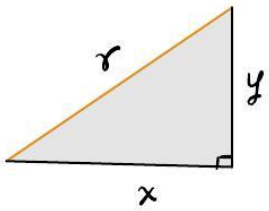
Def Let S be a solid that lies between $x=a$ and $x=b$. If the cross-sectional area of S in the Plane P_x, through x and perpendicular to the x-axis is $A(x)$, where A is a continuous function, then the Volume of S is

$$V = \lim_{n\to\infty} \sum_{i=1}^{n} A(x_i^*)\,\Delta x = \int_a^b A(x)\,dx.$$

Example. Show that the volume of a sphere of radius r is $V = \frac{4}{3}\pi r^3$.

Sln:





by Pythag. thm $y = \sqrt{r^2 - x^2}$.

Rmk: Notice that y is the radious of the red circle so, the cross-sectional area $A(x) = \pi y^2 = \pi\left(\sqrt{r^2-x^2}\right)^2 = \pi(r^2-x^2)$.

Using the definition of the volume with $a=-r$ and $b=r$

$$V = \int_{-r}^{r} A(x)\,dx = \int_{-r}^{r} \pi(r^2-x^2)\,dx$$

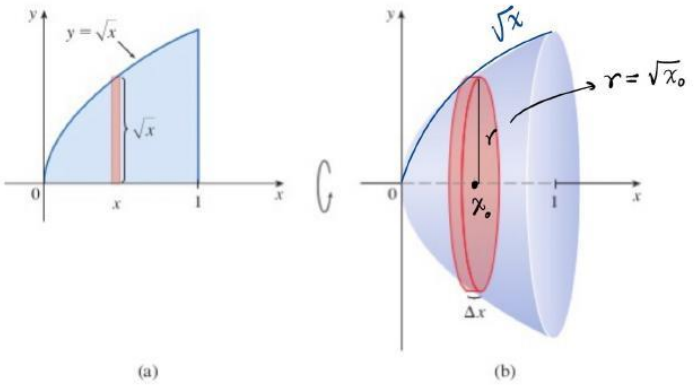
$$= 2\pi\int_0^r (r^2-x^2)\,dx \quad \text{the integral is even.}$$

$$= 2\pi\left(r^2x - \frac{x^3}{3}\Big|_0^r\right) = 2\pi\left(r^3 - \frac{r^3}{3}\right) = 2\pi\left(\frac{2r^3}{3}\right) = \frac{4}{3}\pi r^3.$$

Volumes of solids of Revolution:

Example. Find the volume of the solid obtained by rotating about the x-axis the region under the curve $y=\sqrt{x}$ from 0 to 1.

Sln.



$$A(x_0)=\pi r^2=\pi(\sqrt{x_0})^2=\pi x_0.$$

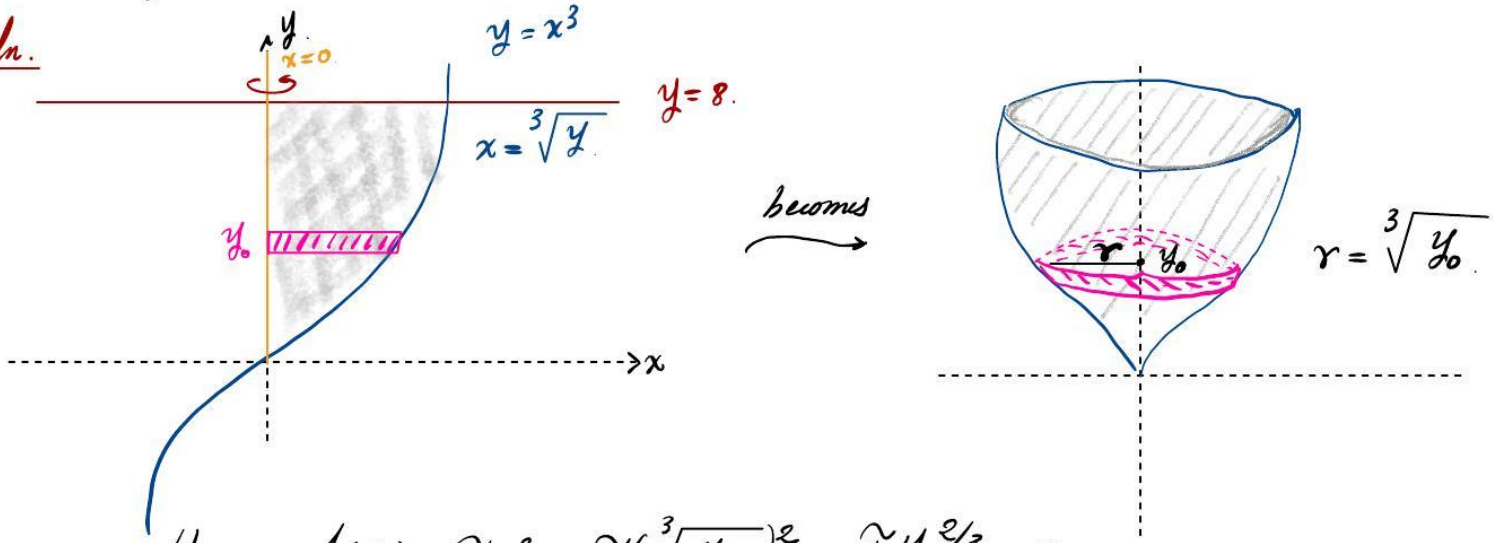
$$A(x)=\pi x.$$

$$V=\int_0^1 A(x)\,dx=\int_0^1 \pi x\,dx$$

$$V=\frac{\pi}{2}x^2\Big|_0^1=\frac{\pi}{2}.$$

Example. Find the volume of the solid obtained by rotating the region bounded by $y=x^3$, $y=8$ and $x=0$ about the y-axis.

Sln.



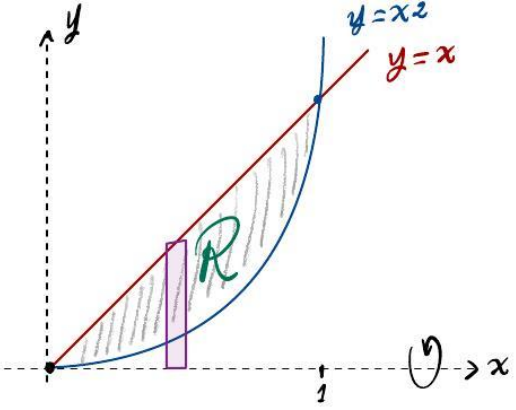
then $A(y_0)=\pi r^2=\pi(\sqrt[3]{y_0})^2=\pi y_0^{2/3}$. So.

$$A(y)=\pi y^{2/3}.$$ Using def of volume.

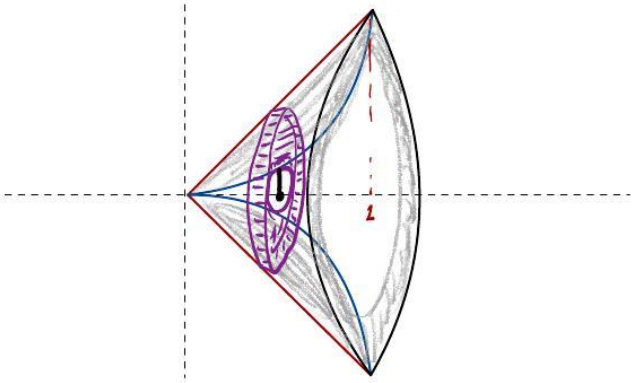
$$V=\int_0^8 A(y)\,dy=\int_0^8 \pi y^{2/3}\,dy=\pi\left(\frac{3}{5}y^{5/3}\Big|_0^8\right)=\frac{96}{5}\pi.$$

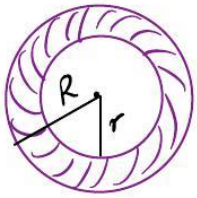
Example. The Region R enclosed by the curves $y=x$ and $y=x^2$ is rotated about the x-axis. Find the volumen of the resulting solid.

Sln.



Solving $x = x^2 \Rightarrow x(x-1)=0 \Rightarrow x=0$
$x=1$.





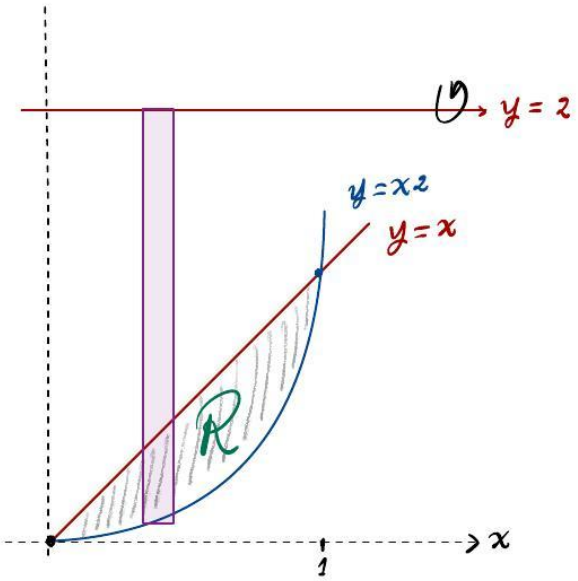
$A = A_B - A_S$ $\quad r = x^2$, $R = x$

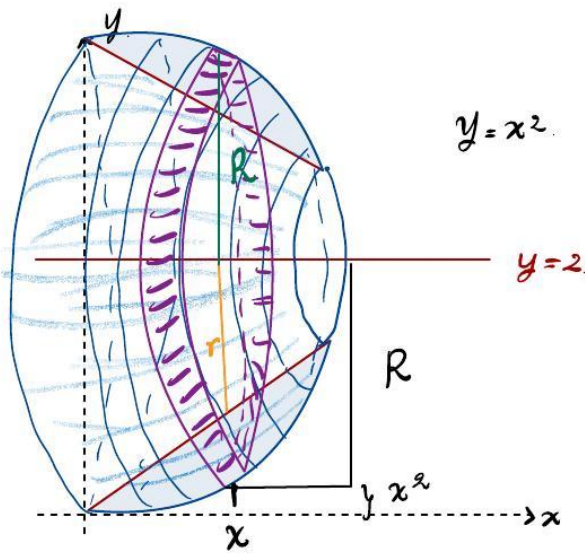
$$A = \pi x^2 - \pi (x^2)^2 = \pi (x^2 - x^4).$$

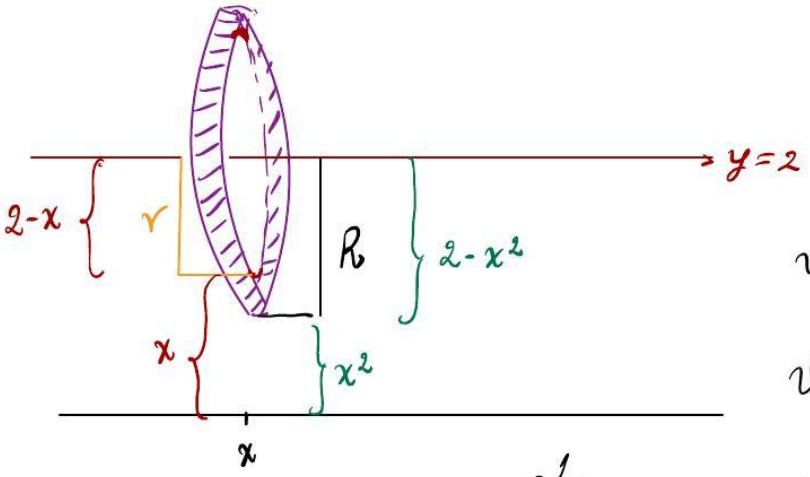
$$v = \int_0^1 A(x)\,dx = \int_0^1 \pi (x^2 - x^4)\,dx = \pi \left(\frac{x^3}{3} - \frac{x^5}{5} \Big|_0^1 \right)$$

$$v = \pi \left(\frac{1}{3} - \frac{1}{5} \right) = \frac{2\pi}{15}.$$

Example. Find the volume of the solid obtained by rotating the region in the previous example about the line $y=2$.







$$A = \pi (R^2 - r^2).$$

$$A(x) = \pi\left((2-x^2)^2 - (2-x)^2\right)$$

$$v = \int_0^1 A(x)\,dx$$

$$v = \int_0^1 \left\{\pi\left((2-x^2)^2 - (2-x)^2\right)\right\} dx$$

$$v = \pi \int_0^1 \left\{4 - 4x^2 + x^4 - (4 - 4x + x^2)\right\} dx$$

$$v = \pi \int_0^1 \left\{4 - 4x^2 + x^4 - 4 + 4x - x^2\right\} dx$$

$$v = \pi \int_0^1 \left\{x^4 - 5x^2 + 4x\right\} dx$$

$$v = \pi \left\{ \frac{x^5}{5} - \frac{5}{3}x^3 + 2x^2 \Big|_0^1 \right\} = \frac{8\pi}{15}.$$